Unit: 6 Unit 5: Linear Equations
Topic: Perpendicular Lines

Name: ________________
Date: __________

Advanced (Habanero)

Q1. What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0

Q2. Which of the following lines is perpendicular to $y = -3x - 2$?
(A) $y = \frac{1}{3}x + 1$
(B) $y = -\frac{1}{3}x - 1$
(C) $y = 3x - 1$
(D) $y = -3x + 1$
(E) $y = \frac{1}{3}x - 1$

Q3. If $y = mx + b$ is perpendicular to $y = 2x - 1$, what is the value of m?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0

Q4. The equation of a line perpendicular to $y = x - 2$ is:
(A) $y = -x + 1$
(B) $y = x + 1$
(C) $y = -x - 1$
(D) $y = x - 1$
(E) $y = -x + 2$

Q5. Which two lines are perpendicular?
(A) $y = 2x + 1$ and $y = -\frac{1}{2}x - 1$
(B) $y = -2x - 1$ and $y = \frac{1}{2}x + 1$
(C) $y = x - 1$ and $y = -x + 1$
(D) $y = x + 1$ and $y = -x - 1$
(E) $y = 2x - 1$ and $y = \frac{1}{2}x + 1$

Q6. If two lines have slopes m_1 and m_2, and $m_1 \cdot m_2 = -1$, the lines are:
(A) parallel
(B) perpendicular
(C) skew
(D) concave
(E) convex

Q7. What is the equation of the line that passes through $(1, 2)$ and is perpendicular to $y = 3x - 1$?
(A) $y = -\frac{1}{3}x + \frac{7}{3}$
(B) $y = \frac{1}{3}x + \frac{5}{3}$
(C) $y = -\frac{1}{3}x + \frac{5}{3}$
(D) $y = \frac{1}{3}x - \frac{1}{3}$
(E) $y = -\frac{1}{3}x - \frac{1}{3}$

Q8. If $y = mx + b$ is perpendicular to the line $x = 2$, what is the value of m?
(A) 0
(B) ∞
(C) $-\infty$
(D) 1
(E) -1

Open Ended Questions (FRQ)

Q9. Prove that the lines $y = 2x + 1$ and $y = -\frac{1}{2}x - 1$ are perpendicular.

Q10. Find the equation of the line that passes through $(2, 3)$ and is perpendicular to the line $y = x - 2$. Then, find the point of intersection of these two lines.

Chef's Tips

- Tip 1: Make sure to check for any calculation errors.
- Tip 2: Provide clear and concise work for the free response questions.
- Tip 3: Use the given equations to find the relationship between the lines.